

Django URL Parameters Guide

A Complete Step-by-Step Tutorial (Hinglish)

Is guide mein Django URL parameters ko 3 alag-alag tareeqon (path, re_path, aur `**kwargs`) se handle karna detail mein sikhaya gaya hai. Yeh notes Mohit Decodes ke tutorial par aadharit hain.

Step 1: Project Aur App Setup Karna

Sabse pehle humein ek naya Django project aur uske andar ek app banani hogi.

- Terminal mein project create karne ke liye command likhein:

```
django-admin startproject myproject2
```

- Project folder ke andar jane ke liye:

```
cd myproject2
```

- Ek nayi application banane ke liye:

```
python3 manage.py startapp blog
```

- **Zaroori Step:** Apne project ki `settings.py` file open karein aur `INSTALLED_APPS` list mein apni nayi app `'blog'` ko add zaroor karein taaki aage chal ke koi error na aaye.

Step 2: path Ke Saath Single Parameters Use Karna

path Django ka ek simple URL mapping function hai jisme hum variables ka data type bhi bata sakte hain (jaise `int` ya `str`).

1. Views banana (views.py)

Sabse pehle `blog/views.py` mein jakar apna view function banayein aur `HttpResponse` import karein. Hum yahan do views bana rahe hain: ek numeric ID ke liye aur ek string username ke liye.

```
from django.http import HttpResponse def post_detail(request, post_id): return  
HttpResponse(f"<h1>Show blog post {post_id}</h1>") def user_profile(request,  
username): return HttpResponse(f"<h1>Profile of {username}</h1>")
```

2. URLs map karna (urls.py)

Ek `urls.py` file banayein (blog folder ke andar) aur unhe connect karein:

```
from django.urls import path from . import views urlpatterns = [ path('post/  
<int:post_id>/', views.post_detail, name='post_detail'), path('user/  
<str:username>/', views.user_profile, name='user_profile'), ]
```

3. Main Project se connect karna

Uske baad main project (myproject2) ke `urls.py` mein `include` import karke apni blog app ke URLs ko main path mein connect karein.

```
from django.contrib import admin from django.urls import path, include urlpatterns  
= [ path('admin/', admin.site.urls), path('blog/', include('blog.urls')), ]
```

Ab aap server run karke (python3 manage.py runserver) URLs ko test kar sakte hain (jaise localhost:8000/blog/post/5).

Step 3: Complex Patterns Ke Liye re_path (Regex)

Kabhi-kabhi URL patterns thode complex hote hain (jaise agar humein sirf 4-digit numbers hi year ki form mein accept karne hain) jo simple path se handle nahi hote. Aise cases mein Regular Expressions (Regex) use hote hain.

`re_path` ko Django URLs se import karein aur pattern banayein:

```
from django.urls import re_path from . import views urlpatterns = [  
re_path(r'^article/(?P<year>[0-9]{4})/$', views.article_by_year), ]
```

Yahan `[0-9]{4}` ka matlab hai ki 0 se 9 ke beech ki exactly 4 digits hi accept hongi. Agar koi browser mein `article/20247` (5 digits) daalega, toh usko URL not found ka error milega.

Step 4: Multiple Parameters Handle Karna

Aap chahein toh ek hi path ke andar ek se zyada dynamic parameters bhi paas kar sakte hain.

```
# urls.py path('article/<int:year>/<str:month>/', views.article_detail) # views.py
def article_detail(request, year, month): return HttpResponse(f"Article for
{month}/{year}")
```

Step 5: **kwargs Ka Use (Best Approach)

Jab URLs mein parameters ki ginti badhti rehti hai (jaise year, month, aur days), toh baar-baar view function mein naye parameters add karna ek accha approach nahi hai. Iska solution Python ka core feature **kwargs (Keyword Arguments) hai.

Isme aapko view mein alag-alag variable lene ki zaroorat nahi hai. View ko simply is tarah likhein:

```
def article_details(request, **kwargs): # kwargs will be a dictionary containing
all URL parameters return HttpResponse(f"Data: {kwargs}")
```

Step-by-Step Dry Run (**kwargs ke sath)

- **Action:** User ne apne browser mein URL type kiya: localhost:8000/blog/article/2024/02/30/.
- **URL Parsing:** Django ke routing system ne match kiya ki request mein 3 dynamic variables hain (year, month, days).
- **Data Flow:** Yeh teeno variables seedha function call mein as keyword arguments (kwargs) paas honge.
- **View Execution:** Aapke function mein parameters aise receive hongii:
kwargs = {'year': 2024, 'month': '02', 'days': 30}
- **Output:** HTTP Response directly is dictionary ko show kar dega bina kisi extra coding ke, jisse multiple parameters ek baar mein hi automatically handle ho gaye.